

KING FAHD UNIVERSITY OF PETROLEUM & MINERALS

KFUPM Business School (KBS) · Department of Finance

The Disappearing Dividend

*155 years of US corporate payout policy, 1871–2023
and what the Gulf model does differently*

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COURSE

Financial Policy

INSTRUCTOR

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DATA

1,830 months · 466 firms

OBSERVATIONS

~21,000 real data points

THE OBSERVATION THAT STARTED THIS

In 2008–09 S&P earnings fell 79%.

The dividend fell 4%.

A century earlier, in the Great Depression, earnings fell 72% and the dividend fell 49%. It did roughly two thirds of the work. Somewhere between those two crises the dividend stopped being a share of profit.

The data

All public, all real, all downloadable — nothing estimated or simulated

1,830

Monthly observations

Shiller series, Jan-1871 to Jun-2023

155

Years covered

a complete history of the US equity market

466

Firms in the cross-section

current S&P 500, matched on ticker

~21,000

Real data points in total

across five public datasets

Dataset	What it gives	Period
Shiller S&P 500 series	Price, dividend, earnings, CPI, 10-year rate, CAPE — monthly	1871–2023
S&P 500 constituents	GICS sector, sub-industry, date added, year founded	current
S&P 500 financials	Per-firm price, P/E, P/B, dividend yield, market cap	current
US 10-year Treasury	Constant maturity yield, monthly	1953–2026
Saudi Aramco results	Free cash flow and distributions, FY2019–FY2025	2019–2025

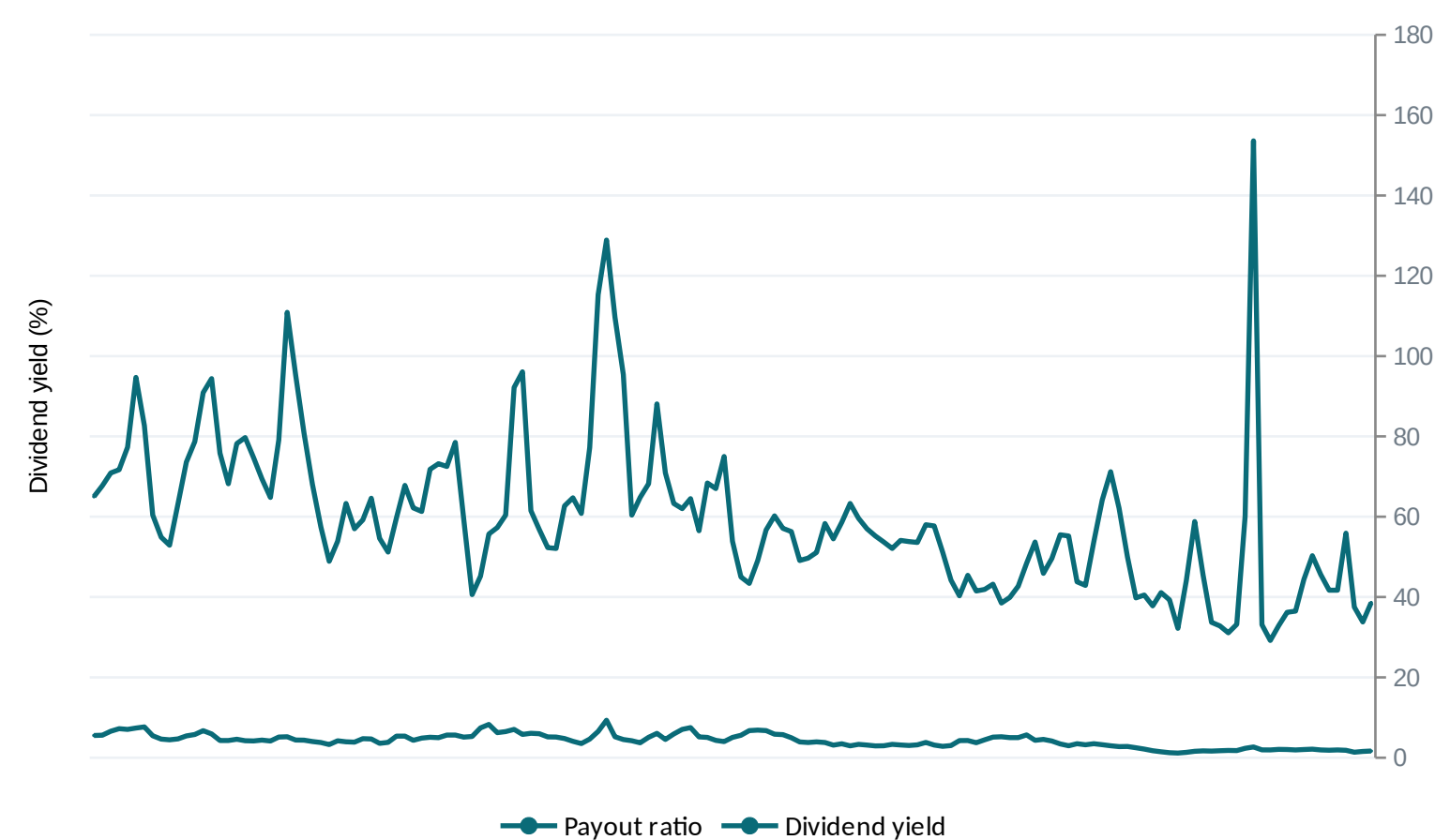
Every source is a public URL listed in the model. The whole analysis is reproducible from scratch.

PART ONE

What happened

155 years of payout policy

Annual payout ratio and dividend yield, 1871–2023



71.7%

Payout ratio, 1871–1900

median of 360 months

40.2%

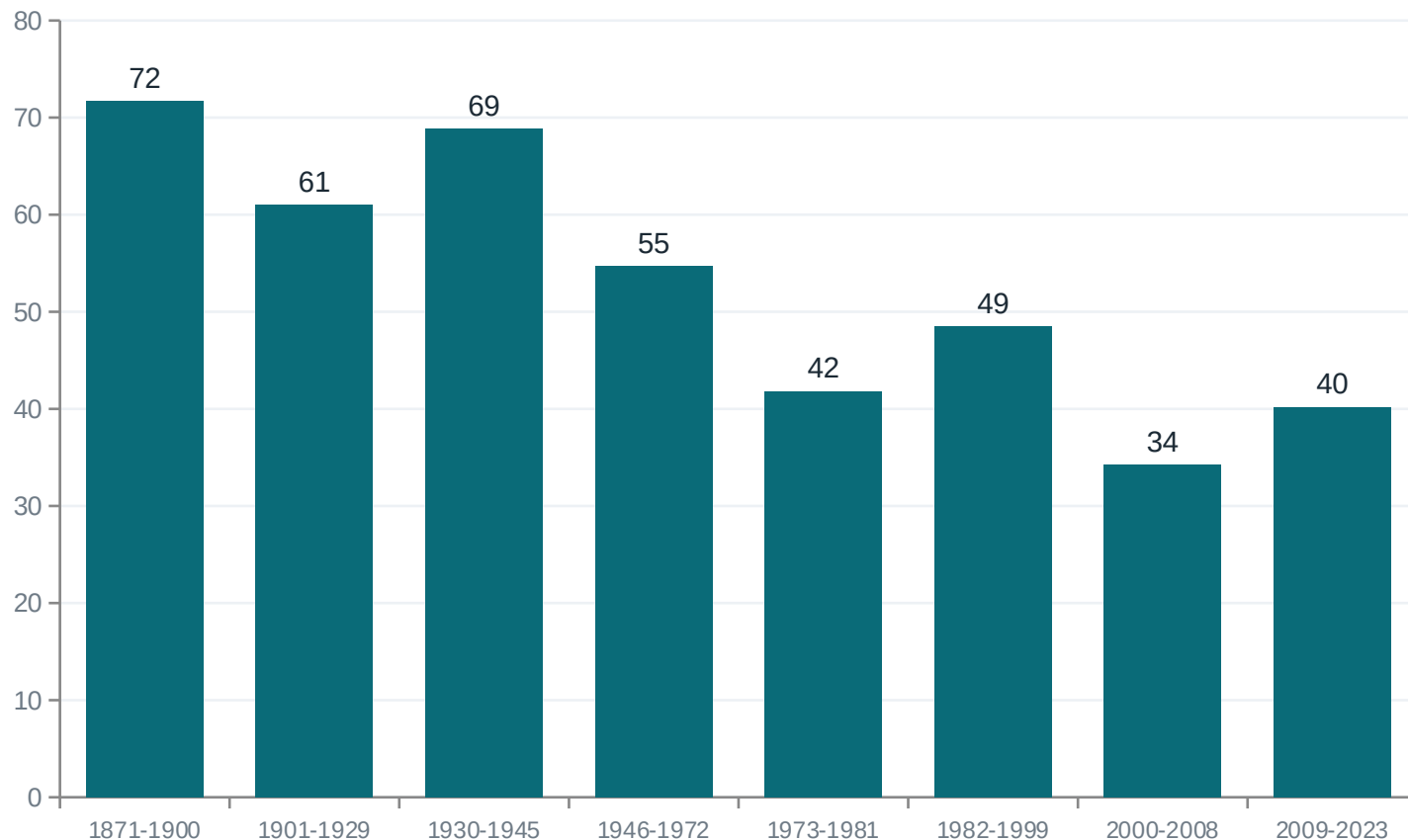
Payout ratio, 2009–2023

median of 174 months

The yield fell further than the payout ratio, because valuations rose at the same time. CAPE went from 17 to 27.

Eight eras

Median payout ratio, dividend yield and smoothing ratio



The eras

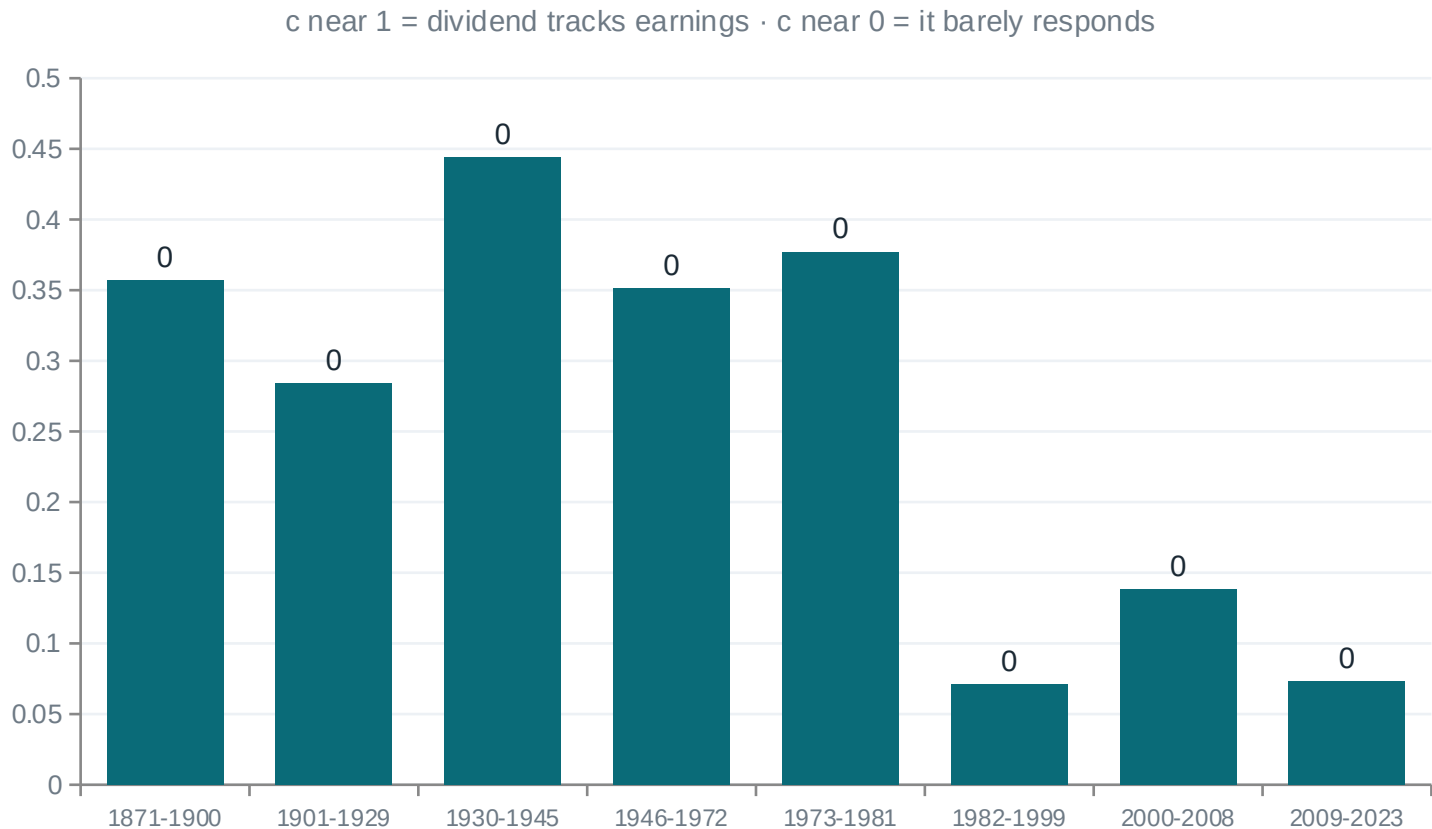
- 1871–1900 Gilded Age — payout 72%, yield 4.8%
- 1901–1929 Pre-Crash — payout 61%
- 1930–1945 Depression & War — payout back to 69% as earnings collapsed
- 1946–1972 Post-war boom — payout 55%
- 1973–1981 Stagflation — payout 42%
- 1982–1999 Buyback era begins — payout 49%, smoothing ratio falls to 0.20
- 2000–2008 — payout 34%, the low point
- 2009–2023 — payout 40%, smoothing ratio 0.09

PART TWO

When did it change?

Lintner's model, estimated in eight eras

$\Delta\text{DPS} = a + c \times (\text{target payout} \times \text{EPS} - \text{lagged DPS})$



0.36

Mean c, 1871–1981

five eras, all significant

0.09

Mean c, 1982–2023

three eras

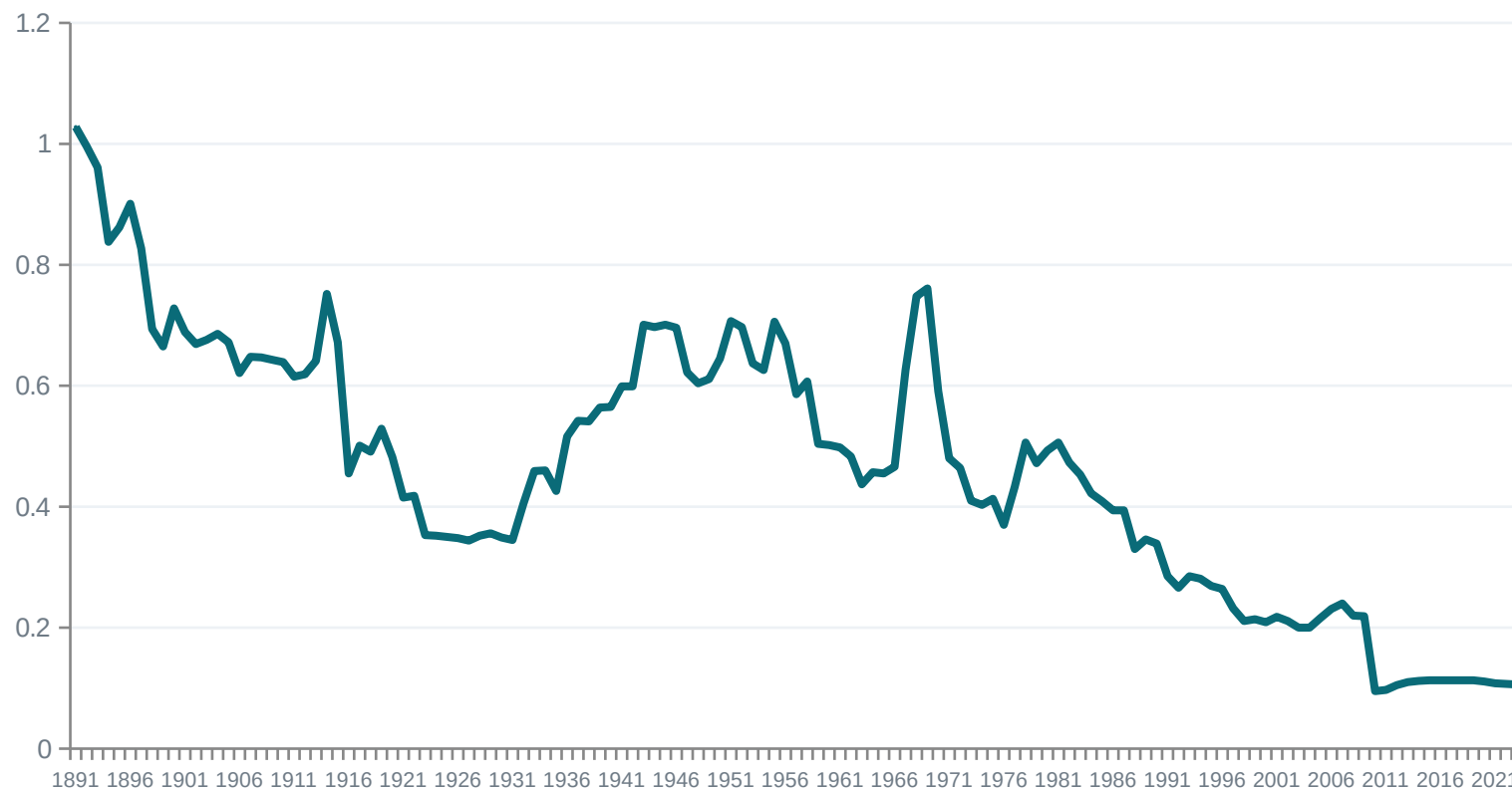
Lintner's own 1956 estimate was about 0.30 — almost exactly what this data gives for every era up to 1981.

Source: Robert Shiller long-run S&P 500 series, 1,830 monthly observations, Jan-1871 to Jun-2023 — regressions estimated on annual data within each era

The break, without assuming where it is

Rolling 20-year smoothing ratio: $\text{sd}(\text{dividend growth}) \div \text{sd}(\text{earnings growth})$

Lower = dividend is smoother relative to earnings



0.817

Smoothing ratio, 1871–1900

dividends moved with earnings

0.089

Smoothing ratio, 2009–2023

they barely move at all

Dividends became 9.2× smoother relative to earnings. The decline is continuous — the era split is presentation, not assumption.

Why 1982?

The legal change that gave managers a second lever

SEC Rule 10b-18

Adopted in 1982, Rule 10b-18 created a safe harbour: a company repurchasing its own shares within specified volume, timing and price conditions would not be treated as manipulating its stock.

Before it, open-market buybacks carried real legal risk and were rare. After it, they became the standard mechanism for returning variable amounts of cash.

The dividend was left to do one job — signal stability. The buyback took the other — distribute whatever is left over.

What the data shows around it

- Speed of adjustment falls from 0.36 to 0.09
- Implied half-life of adjustment rises from 3.4 to 5.9 years
- Smoothing ratio falls from 0.817 to 0.089
- In 2009–2023 the Lintner coefficient has a t-statistic of 0.89 — statistically indistinguishable from zero
- The payout ratio falls, but the collapse in responsiveness is the larger change

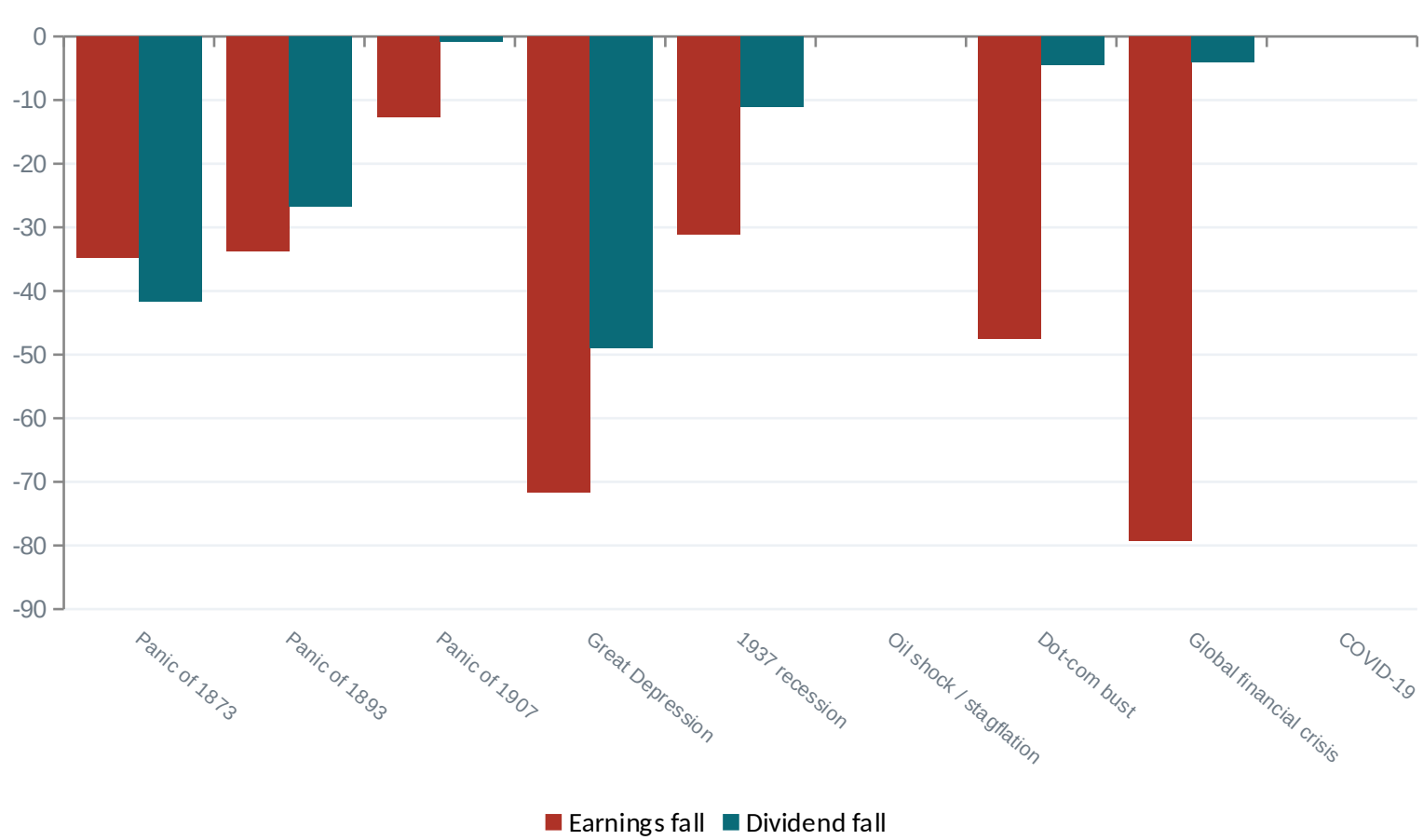
Honest caveat: this project does not have a buyback series. The 1982 interpretation is an inference from timing and from the collapse in responsiveness — not a direct test.

PART THREE

Does it hold up?

Test one: how dividends behave in crises

Peak-to-trough falls in nine episodes, 1873–2020



-72% / -49%

Great Depression

earnings / dividend

-79% / -4%

Global financial crisis

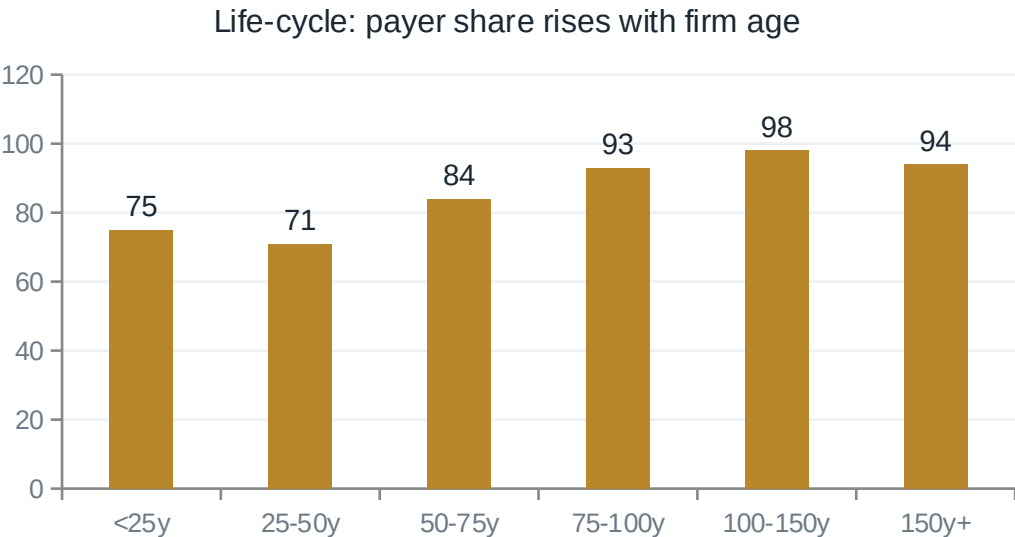
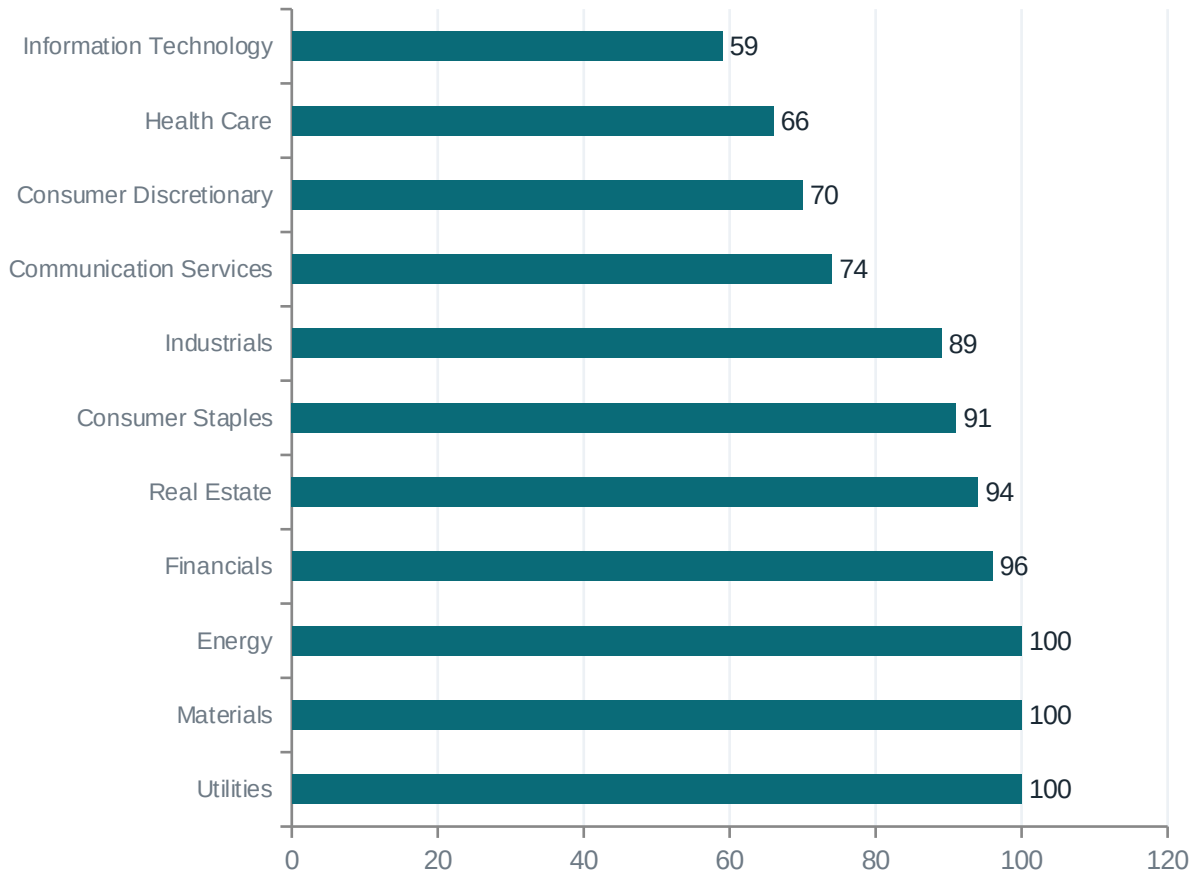
earnings / dividend

In 1929–33 the dividend absorbed two thirds of the shock. In 2007–09 it absorbed almost none.

Source: Robert Shiller long-run S&P 500 series, 1,830 monthly observations, Jan-1871 to Jun-2023

Test two: who still pays

466 current S&P 500 firms, by sector and by age

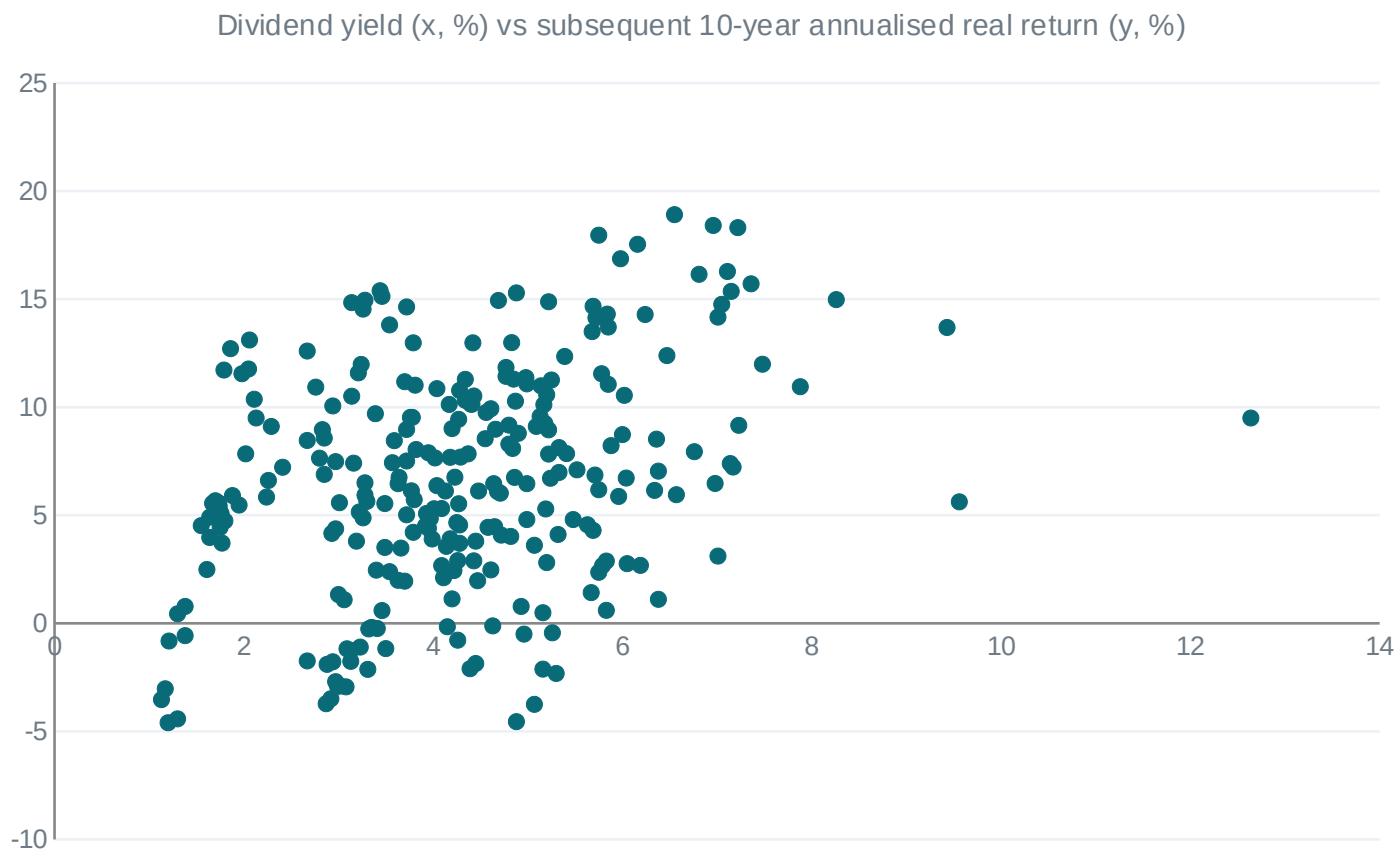


DeAngelo, DeAngelo & Stulz (2006) predict firms begin paying once they mature and run out of investment opportunities. Payer share rises from 75% among firms under 25 to 98% among those aged 100–150. The theory survives contact with the data.

Source: Current S&P 500 constituents and financials — 466 of 503 firms matched on ticker

Test three: does the yield still tell you anything?

Starting dividend yield against the following 10 years of real return



0.124

R² on dividend yield

correlation 0.35

0.283

R² using 1/CAPE

valuation does better

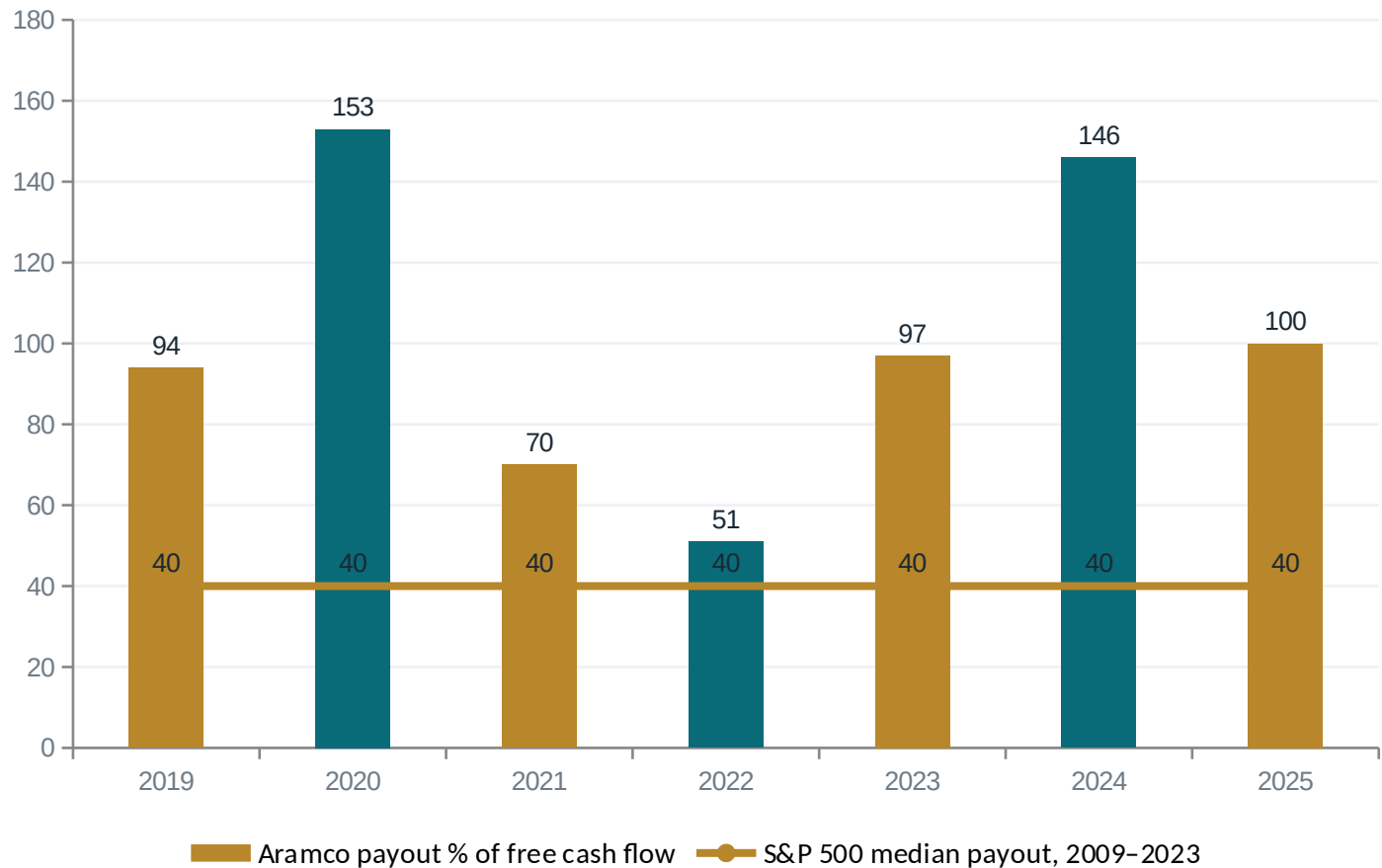
Caveat: 1,590 overlapping windows share 119 of 120 months with their neighbours. The effective sample is nearer 15 than 1,590 — reported as description, not as a signal.

PART FOUR

What about here?

The Gulf model runs the other way

Saudi Aramco payout as a share of free cash flow, FY2019–FY2025



Same instrument, opposite philosophy

Aramco distributed 92% of cumulative free cash flow over seven years — a scale the US index abandoned a century ago.

When it needed flexibility in 2023 it did not introduce a buyback. It introduced a second, variable dividend.

Why? Ownership. A dispersed US shareholder base can sell shares to manufacture income, so a low smoothed dividend plus buybacks is efficient. A sovereign holding 97.5% of the register cannot sell without giving up control, and needs the cash for a budget.

Payout policy is downstream of ownership structure.

Conclusions

Five findings, in order of confidence

- | | | |
|---|---|---|
| 1 | The dividend changed character, not just level | The payout ratio roughly halved, but the collapse in responsiveness to earnings is the larger and more interesting change. |
| 2 | The break is datable to the early 1980s | Speed of adjustment falls from 0.36 to 0.09; smoothing ratio falls 6-fold; rolling windows confirm it without assuming era boundaries. |
| 3 | Lintner was right about what managers want | They want to smooth. What changed after 1982 is that they finally had an instrument — the buyback — that let them smooth completely. |
| 4 | The life-cycle theory survives | Payer share rises monotonically with firm age across the four oldest bands of the modern cross-section. |
| 5 | Payout policy is downstream of ownership | The Gulf contrast makes visible what the US time series alone cannot: the same instrument serves opposite purposes under different registers. |

Limitations

Where this analysis is weakest — stated before you ask

1

Index-level, not firm-level, for the long run

The Shiller series is an aggregate. It cannot separate a fall in payout driven by changing index composition from one driven by individual firms changing policy. Firm-level data back to 1871 does not exist in accessible form.

2

No buyback series — the biggest gap

The 1982 interpretation rests on timing and on the collapse in dividend responsiveness. I have no repurchase data to test it directly. This is the obvious extension and I would start there.

3

Overlapping windows in the predictive regression

1,590 monthly observations of a 10-year forward return share 119 of 120 months with their neighbours. Conventional significance tests would be badly overstated.

4

Era boundaries are chosen, not estimated

The rolling 20-year chart is included precisely so the break can be seen without relying on those choices. A formal structural break test would be stronger.

The dividend did not disappear. It was demoted.

Conclusion

For a century it was the mechanism by which profit reached shareholders — it moved with earnings, it fell in crises, and Lintner could estimate a stable adjustment speed of about 0.3 in every era from 1871 to 1981.

After 1982 it became something else: a small, smoothed, defended payment that barely responds to profit at all. In 2008–09 earnings fell 79% and it fell 4%. **The variable component of shareholder return moved to buybacks, and the dividend was left holding only the signal.**

And none of this is a law of finance. Aramco, with a sovereign holding 97.5% of its register, runs the opposite policy — because payout policy is downstream of who owns the shares.

Thank you

Questions

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